NestSection: Consistent Multi-Chamber Redistricting via Compatible Factorization Spines

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May 2026

Abstract

Congressional, state senate, and state house districts for the same state are currently drawn by three independent redistricting processes. This independence creates structural opportunities for coherent gerrymandering: a partisan actor who controls all three processes can pack or crack a geographic community at every legislative level simultaneously. *NestSection* eliminates this opportunity by forcing all three chambers to share a common hierarchical bisection tree—the *compatible factorization spine*.

The spine is determined by $g = \gcd(C, S, H)$, where C, S, and H are the congressional, senate, and house seat counts. When g equals the smallest chamber size, nesting is exact (score = 0); when $g = 1$, the chambers share only a trivial root (score $\approx$ 100). We apply the SpineCompatibility score to all 50 US states under the 2020 apportionment, finding 11 states with score 0 (including Oregon, Alabama, Montana, and Wyoming), and identifying the structural reasons why 39 states require a best-effort Mode 3 nesting.

The central legal argument is that a shared geographic hierarchy is an anti-gerrymandering mechanism by construction: a partisan actor who locks in a congressional spine cannot then draw senate boundaries that contradict it, because the senate tree is geometrically derived from the same trunk sequence. We discuss the statutory design implications and propose a nestability threshold $g \geq 5$ as a practical criterion for mandating exact nesting.

Keywords: redistricting, multi-chamber consistency, factorization spine, gerrymandering, hierarchical bisection, apportionment

Contents

1 Introduction

Every decade following the census, state legislatures draw three sets of district boundaries for the same geographic territory: congressional districts (determined by federal apportionment), state senate districts, and state house districts. Under current practice these three processes are legally and computationally independent. A legislature controlling all three maps can exploit this independence to reinforce partisan advantages across chambers: pack a geographic community into a safe district at the congressional level, pack the same community into a safe senate district, and pack it again at the house level. No single anti-gerrymandering doctrine spans all three chambers simultaneously.

1.1 The Structural Independence Problem

Consider North Carolina after 2020 apportionment: $C = 14$ congressional seats, $S = 50$ senate seats, $H = 120$ house seats. A partisan legislature facing a competitive map at one level can draw the other two levels to compensate. If a court strikes down the congressional map, the partisan sorting of communities is preserved in the senate and house maps, which then serve as the foundation for the next round of congressional redistricting. The maps are legally independent but geographically entangled.

The *structural* version of this problem is even more basic. The three seat counts (C, S, H) rarely share a compatible prime factorization. There is in general no way to recurse a 14-seat bisection tree to produce exactly 50 senate children and 120 house grandchildren. Each chamber's map is drawn by a different algorithm, consultant, or commission, with different geographic anchors, producing three maps that are at best loosely correlated.

1.2 NestSection

We propose *NestSection*, an algorithm that forces the three chambers to share a common *compatible factorization spine*. The spine is the coarsest geographic division that all three chambers agree on: it is determined by $g = \gcd(C, S, H)$, the greatest common divisor of the three seat counts. Within each of the g spine regions, the three chambers recurse independently, but they share the same geographic boundaries at the top level.

The key insight is that g determines how much cross-chamber structure can be shared. When $g = \min(C, S, H)$, the smallest chamber's seat count divides both larger counts exactly, and the nesting is perfect: every district of the smaller chamber is a union of districts from the larger chambers. When $g = 1$, no non-trivial common structure exists and the three chambers can only share a root node (the state boundary itself).

Formal compatibility score. We define the *spine compatibility score* as:

$$\sigma(C, S, H) \;=\; \left(1 - \frac{\gcd(C, S, H)}{\min(C, S, H)}\right) \times 100 \;\in\; [0, 100]. \tag{1}$$

A score of 0 means the smallest chamber is entirely captured by the shared trunk; a score of 100 means the trunk is trivially a single root.

Main contributions.

1. A formal definition of compatible factorization spines and a polynomial-time algorithm (spine^*) for their construction (Section 3).
2. A 50-state compatibility census under the 2020 apportionment, including an exact classification of which states admit strict nesting and which require best-effort Mode 3 nesting (Section 4).
3. A legal argument that shared-spine redistricting is an anti-gerrymandering mechanism by construction, and a statutory design proposal for states where $g \geq 5$ (Section 5).

Roadmap. Section 2 reviews multi-member district theory, nested districting proposals, and the gerrymandering literature relevant to cross-chamber coherence. Section 3 defines the spine^* algorithm formally, proves its key properties, and describes the three nesting modes. Section 4 presents the 50-state compatibility census and three case studies (Oregon, North Carolina, Texas). Section 5 develops the legal argument, addresses limitations, and outlines future work. Section 6 summarizes.

Implementation note. The spine^* algorithm is implemented in the `redist-apportion` crate of the `bisect` Rust workspace; the `compatible_spines`, `spine_compatibility_score`, and `us_state_compatibility_table` functions are publicly exported and tested against all 50 states.

2 Related Work

2.1 Multi-Member Districts and Nested Structures

Multi-member district (MMD) systems are well-studied in political science and electoral law. Grofman [1993] reviews proportional and semi-proportional MMD systems and their effects on minority representation; Bowler and Farrell [1999] analyzes strategic voting under MMD arrangements. Neither line of work addresses the computational question of constructing hierarchically consistent maps across chambers.

Nested electoral structures—where one legislative body's districts are unions of another's—appear in several European parliaments and in some US state constitutions. The classic example is the German Bundestag, which pairs single-member constituencies with proportional list seats drawn from the same geographic regions. Taagepera and Shugart [1989] analyzes the seat-product model for mixed-member systems; Cox [2006] examines how nesting affects strategic entry. These works treat nesting as a political science fact, not a computational design problem.

The US context is different: the three chambers are separately apportioned, separately litigated, and separately optimized. No prior computational redistricting literature, to our knowledge, treats consistency across chambers as an algorithmic objective.

2.2 Computational Redistricting

The algorithmic redistricting literature largely focuses on single-chamber problems. Altman and McDonald [2011] and Fifield et al. [2020] describe Markov chain Monte Carlo (MCMC) sampling over single-chamber district plans; Chen and Rodden [2013] uses simulated annealing to generate unbiased district plans for gerrymandering detection. Graph-partitioning approaches based on METIS [Karypis and Kumar, 1998] are widely used in practice; they optimize edge-cut as a proxy for compactness.

Recursive bisection methods [Simon, 1991] produce hierarchical district trees naturally, but prior work applies them to a single chamber at a time. The GeoSection algorithm [Della-Libera, 2026b] extends recursive bisection with ratio-optimal METIS partitioning; ApportionRegions [Della-Libera, 2026a] provides integer seat allocation for spine regions. NestSection builds directly on both.

The GerryChain project [DeFord et al., 2021] and related ensemble methods focus on detecting gerrymandering by sampling the space of valid plans. NestSection is complementary: rather than detecting gerrymandering post hoc, it constrains the feasible plan space by design. The compatibility score σ determines how tightly the constraint binds.

2.3 Cross-Chamber Gerrymandering

Gelman and King [1994] note that partisan effects compound across legislative chambers when maps are drawn by the same partisan majority. Chen and Cottrell [2016] documents cases where congressional and state legislative maps were drawn sequentially to reinforce each other's partisan effects in North Carolina, Wisconsin, and Pennsylvania.

The legal literature on cross-chamber coherence is sparse. *Rucho v. Common Cause*, 588 U.S. 684 (2019), held that federal courts cannot police partisan gerrymandering in congressional maps, but left state courts free to act under state constitutional provisions [Miller, 2020]. Several state supreme courts have struck down single-chamber maps (Pennsylvania *LWV v. Commonwealth* (2018); North Carolina *Harper v. Hall* (2022); New York *Harkenrider v. Hochul* (2022)) without addressing the cross-chamber consistency problem.

Allen v. Milligan, 599 U.S. 1 (2023), reaffirmed the applicability of Section 2 of the Voting Rights Act (VRA) to single-member congressional districts, but again in a single-chamber context. *Callais v. Landry* (W.D. La. 2024) addressed VRA compliance in a state legislative map; the court's focus on proportionality at a single level highlights the gap that NestSection addresses: if VRA compliance is evaluated chamber by chamber, a partisan actor can satisfy each chamber's requirements independently while maintaining coherent cross-chamber advantages.

2.4 Factorization and Apportionment Theory

The connection between prime factorization and hierarchical bisection trees appears in Della-Libera [2026a] (ApportionRegions) and implicitly in the METIS partitioning literature. Huntington [1921] and Balinski and Young [1982] develop the theory of divisor methods for apportionment; neither addresses multi-chamber compatibility. The NestSection compatibility score $\sigma(C, S, H) = (1 - g/m) \times 100$ is, to our knowledge, a novel metric for cross-chamber structural alignment.

Divisibility conditions similar to our strict-nesting requirement appear in local government law: many state constitutions require county-level districts to be composed of whole precinct units, and precinct units to be composed of whole census blocks. NestSection extends this hierarchy upward to the legislative level.

3 The Spine* Algorithm

3.1 Preliminaries

Let $G = (V, E, w)$ be a planar graph representing the census units (tracts or blocks) of a state, with vertex weights w_v equal to the 2020 census population of unit v. Let C, S, and H be the state's congressional, senate, and house seat counts under the 2020 apportionment.

A *bisection spine* for a chamber with k seats is a sequence of positive integers $[s_1, s_2, \ldots, s_\ell]$ with $\prod_i s_i = k$, interpreted as a recursion schedule: first split G into s_1 sub-graphs, then each sub-graph into s_2 sub-sub-graphs, and so on. The canonical choice is the prime factorization of k in non-decreasing order.

Definition 1 (Compatible Spines)**.** *A* compatible triple of spines *for* (C, S, H) *is a triple* (P_C, P_S, P_H) *of bisection spines such that* P_C*,* P_S*, and* P_H *share a common prefix* τ*, called the* trunk*. The trunk product* $\prod \tau$ *equals* $g = \gcd(C, S, H)$*.*

The trunk τ is the prime factorization of g in non-decreasing order. The chamber-specific tails are the prime factorizations of C/g, S/g, and H/g respectively. A tail of length zero (quotient $= 1$) means the chamber ends exactly at the trunk level.

3.2 Algorithm: CompatibleSpines

Algorithm 1 $\textsc{CompatibleSpines}(C, S, H)$

Require: $C, S, H \geq 1$
Ensure: (P_C, P_S, P_H) — compatible triple of bisection spines

```
g ← gcd(C, S, H)
τ ← PrimeFactors(g)                    ▷ sorted non-decreasing
tail_C ← PrimeFactors(C/g)
tail_S ← PrimeFactors(S/g)
tail_H ← PrimeFactors(H/g)
P_C ← τ‖tail_C
P_S ← τ‖tail_S
P_H ← τ‖tail_H
return (P_C, P_S, P_H)
```

Proposition 2 (Correctness)**.** *For any* $C, S, H \geq 1$*,* $\textsc{CompatibleSpines}(C, S, H)$ *returns a triple* (P_C, P_S, P_H) *satisfying:*

1. *$\prod P_C = C$, $\prod P_S = S$, $\prod P_H = H$.*

2. *P_C, P_S, P_H all begin with $\tau = \textsc{PrimeFactors}(\gcd(C, S, H))$.*

3. *The trunk τ is the* longest *common prefix that can be guaranteed for any (C, S, H) with the given GCD.*

Proof. (1) The product of a spine equals the product of its trunk and tail. $\prod \tau = g$ and $\prod \mathrm{tail}_C = C/g$, so $\prod P_C = g \cdot (C/g) = C$. Analogously for S and H.

(2) By construction, all three spines begin with τ.

(3) Suppose a longer common prefix τ' with $\prod \tau' = g' > g$ existed. Then g' would divide all of C, S, H, contradicting $g = \gcd(C, S, H)$. □

3.3 Compatibility Score

Definition 3 (Spine Compatibility Score)**.** *The* spine compatibility score *of (C, S, H) is:*

$$\sigma(C, S, H) \;=\; \left(1 - \frac{\gcd(C, S, H)}{\min(C, S, H)}\right) \times 100. \tag{2}$$

Lemma 4 (Score Properties)**.** *1. $\sigma(C, S, H) \in [0, 100]$.*

2. $\sigma(C, S, H) = 0$ if and only if $\gcd(C, S, H) = \min(C, S, H)$, i.e., the smallest chamber's count divides both larger counts.

3. $\sigma(C, S, H) = 100$ only in the degenerate limit; the maximum attained score for any US state is ≈ 96.8 (Texas: $g = 1$, $\min = 31$).

Proof. (1) Since $\gcd(C, S, H) \leq \min(C, S, H)$, the fraction is in $[0, 1]$. (2) Immediate from the definition. (3) The score equals $100(1 - 1/m)$ when $g = 1$. For any US state, $\min(C, S, H) \leq C \leq 52$, so the score is at most $100(1 - 1/52) \approx 98.1$, but no state achieves this because $C = 52$ (California) has $g = 4 > 1$. □

Theorem 5 (Bimodality Gap)**.** *For positive integers C, S, H with $g = \gcd(C, S, H)$ and $m = \min(C, S, H)$, the compatibility score $\sigma = (1 - g/m) \times 100$ satisfies*

$$\sigma \;\in\; \{0\} \;\cup\; [50, 100).$$

There are no states with $0 < \sigma < 50$.

Proof. Suppose for contradiction that $0 < \sigma < 50$. Then

$$0 \;<\; 1 - \frac{g}{m} \;<\; \frac{1}{2}, \quad \text{i.e.,} \quad \frac{m}{2} \;<\; g \;<\; m.$$

Since $g = \gcd(C, S, H)$ divides every element, in particular $g \mid m$. We therefore need a divisor of m that lies strictly between $m/2$ and m. Such a divisor would be m/p for some integer p with $1 < p < 2$ — but no such integer exists, since no prime is less than 2.

More precisely: if $g \mid m$ and $g > m/2$, then $m/g < 2$, so $m/g = 1$ (it must be a positive integer), which gives $g = m$ and $\sigma = 0$ — a contradiction with $\sigma > 0$.

Boundary cases.

- If $m = 1$, then $g = \gcd(C, S, H) = 1$ (the only positive divisor of 1 is 1 itself), so $\sigma = (1 - 1/1) \times 100 = 0$. Consistent.
- The infimum of the non-zero range is achieved when $g = 1$ and $m = 2$: $\sigma = (1 - 1/2) \times 100 = 50$. Hence the non-zero values begin at exactly 50, confirming $\sigma \in \{0\} \cup [50, 100)$.

□

Remark 6. The Bimodality Gap theorem is a pure number-theoretic fact: it holds for any positive integers (C, S, H), not merely for current US apportionments. The empirical observation that no US state falls in the range $0 < \sigma < 50$ is therefore not an accident of the current apportionment cycle but a structural invariant of the GCD-based score.

3.4 Nesting Modes

The compatibility score determines which nesting mode is achievable.

Mode 1 — Strict nesting ($\sigma = 0$). When $\gcd(C, S, H) = \min(C, S, H)$, the smallest chamber is exactly the trunk. Its districts are the trunk regions. The larger chambers are refinements of the trunk. No boundary violations occur.

Mode 2 — Partial nesting ($0 < \sigma < 50$). *Mode 2 is mathematically empty*: Theorem 5 proves that no positive integer triple (C, S, H) can produce a score in $(0, 50)$. The mode is defined here for completeness and to establish the classification vocabulary, but it cannot arise for any real apportionment. The trunk covers more than half the structural degrees of freedom of the smallest chamber. The spine provides a strong geographic anchor, but the tail splits produce boundary zones where chambers partially overlap.

Mode 3 — Best-effort nesting ($\sigma \geq 50$). The trunk is a small fraction of the smallest chamber. The chambers can share only a coarse geographic skeleton. A mismatch tolerance τ_{pop} (population fraction allowed in boundary zones) is required.

3.5 The Full NestSection Algorithm

NestRefine. NESTREFINE splits each senate district into h_i/s_i house sub-districts. When h_i/s_i is not an integer, it assigns floor or ceiling counts by population weight and marks boundary zones where a house district straddles a senate boundary. The population fraction in boundary zones must not exceed τ_{pop}.

Remark 7 (Uniformity in Mode 1 states). In Mode 1 states ($g = C$, i.e. $\sigma = 0$), every trunk region receives exactly $c_i = 1$ congressional seat: since $C/g = C/C = 1$, the APPORTIONREGIONS call assigns exactly one seat to each trunk region without any inter-region rounding. Population heterogeneity across trunk regions is handled by GEOSECTION's ratio-optimal bisection within each region. The senate seats $s_i = S/g$ and house seats $h_i = H/g$ are likewise integers (since $g = \gcd(C, S, H)$ divides S and H), so $h_i/s_i = H/S$ is a uniform integer quotient within each trunk region. For Oregon ($g = 6$: $H/S = 60/30 = 2$) and Alabama

Algorithm 2 $\textsc{NestSection}(G, C, S, H, \tau_{\text{pop}})$

Require: Graph G, seat counts C, S, H, tolerance $\tau_{\text{pop}} \geq 0$
Ensure: Tri-chamber district plan (D_C, D_S, D_H)

$(P_C, P_S, P_H) \leftarrow \textsc{CompatibleSpines}(C, S, H)$
$g \leftarrow \gcd(C, S, H)$
$\mathcal{R} \leftarrow \textsc{GeoSection}(G, g)$ ▷ g trunk regions via GeoSection [Della-Libera, 2026b]
for all trunk region $R_i \in \mathcal{R}$, $i = 1, \ldots, g$ **do**
  $c_i \leftarrow \textsc{ApportionRegions}(\{w(R_j)\}_j, C)[i]$ ▷ Integer seats per trunk [Della-Libera, 2026a]
  $s_i \leftarrow S/g$
  $h_i \leftarrow H/g$
  $D_C[i] \leftarrow \textsc{GeoSection}(R_i, c_i)$
  $D_S[i] \leftarrow \textsc{GeoSection}(R_i, s_i)$
  $D_H[i] \leftarrow \textsc{NestRefine}(D_S[i], h_i/s_i, \tau_{\text{pop}})$
end for
return $(\bigcup_i D_C[i],\ \bigcup_i D_S[i],\ \bigcup_i D_H[i])$

($g = 7$: $H/S = 105/35 = 3$), $\textsc{NestRefine}$ reduces to clean recursive $\textsc{GeoSection}$ calls with no boundary zones.

3.6 Trunk Size and Geographic Scale

The trunk size g determines the geographic scale of the shared structure. For Oregon ($g = 6$), the shared trunk has 6 regions averaging roughly 700,000 residents each — approximately the size of a single congressional district. For Alabama ($g = 7$), the trunk has 7 regions matching the 7 congressional districts exactly, and senate districts are unions of exactly 5 trunk regions each.

For states with $g = 1$ (e.g., Texas with $C = 38$, $S = 31$, $H = 150$), the "trunk" is the entire state, and the three chambers share no sub-state geographic anchor beyond the state boundary. In these cases NestSection reduces to three independent GeoSection calls with the shared property that they start from the same census graph G — a weak form of consistency.

3.7 Complexity

Algorithm 1 runs in $O(\sqrt{\max(C, S, H)})$ time: GCD computation by Euclidean algorithm runs in $O(\log)$, but prime factorization by trial division runs in $O(\sqrt{n})$, and for seat counts $n \leq 200$ this distinction is practically irrelevant. The dominant cost of Algorithm 2 is the $g + 3$ calls to $\textsc{GeoSection}$, each of which runs in time proportional to graph-partitioning cost. The total additional cost over three independent GeoSection runs is the g trunk-level calls, which is at most $\min(C, S, H)$ additional partitioning operations.

4 Evaluation: 50-State Compatibility Census

We apply the SpineCompatibility score to all 50 US states using the 2020 apportionment congressional seat counts and the 2020 state legislative chamber sizes from the National Conference of State Legislatures (NCSL).

4.1 Full Compatibility Table

Table 1 reports all 50 states sorted by compatibility score σ (ascending). States are classified into three groups: *Strictly compatible* ($\sigma = 0$, $g = \min(C, S, H)$), *Partially compatible* ($0 < \sigma < 50$, strong trunk), and *Structurally incompatible* ($\sigma \geq 50$, weak trunk).

Table 1: NestSection compatibility scores for all 50 states (2020 apportionment). C = congressional seats, S = state senate seats, H = state house seats, $g = \gcd(C, S, H)$, σ = spine compatibility score.

State	Code	C	S	H	g	σ	Class
Alaska	AK	1	20	40	1	0.00	Strict
Alabama	AL	7	35	105	7	0.00	Strict
Delaware	DE	1	21	41	1	0.00	Strict
Montana	MT	2	50	100	2	0.00	Strict
North Dakota	ND	1	47	94	1	0.00	Strict
New Hampshire	NH	2	24	400	2	0.00	Strict
Oregon	OR	6	30	60	6	0.00	Strict
South Dakota	SD	1	35	70	1	0.00	Strict
Vermont	VT	1	30	150	1	0.00	Strict
West Virginia	WV	2	34	100	2	0.00	Strict
Wyoming	WY	1	30	60	1	0.00	Strict
Hawaii	HI	2	25	51	1	50.00	Incompatible
Iowa	IA	4	50	100	2	50.00	Incompatible
Idaho	ID	2	35	70	1	50.00	Incompatible
Louisiana	LA	6	39	105	3	50.00	Incompatible
Maine	ME	2	35	151	1	50.00	Incompatible
Mississippi	MS	4	52	122	2	50.00	Incompatible
Rhode Island	RI	2	38	75	1	50.00	Incompatible
Arizona	AZ	9	30	60	3	66.67	Incompatible
Kentucky	KY	6	38	100	2	66.67	Incompatible
Nebraska	NE	3	49	49	1	66.67	Incompatible
New Jersey	NJ	12	40	80	4	66.67	Incompatible
New Mexico	NM	3	42	70	1	66.67	Incompatible
Tennessee	TN	9	33	99	3	66.67	Incompatible
Arkansas	AR	4	35	100	1	75.00	Incompatible
Kansas	KS	4	40	125	1	75.00	Incompatible

Table 1 (continued)

State	Code	C	S	H	g	σ	Class
Nevada	NV	4	21	42	1	75.00	Incompatible
Utah	UT	4	29	75	1	75.00	Incompatible
Connecticut	CT	5	36	151	1	80.00	Incompatible
Ohio	OH	15	33	99	3	80.00	Incompatible
Oklahoma	OK	5	48	101	1	80.00	Incompatible
Florida	FL	28	40	120	4	85.71	Incompatible
Georgia	GA	14	56	180	2	85.71	Incompatible
North Carolina	NC	14	50	120	2	85.71	Incompatible
South Carolina	SC	7	46	124	1	85.71	Incompatible
Colorado	CO	8	35	65	1	87.50	Incompatible
Maryland	MD	8	47	141	1	87.50	Incompatible
Minnesota	MN	8	67	134	1	87.50	Incompatible
Missouri	MO	8	34	163	1	87.50	Incompatible
Wisconsin	WI	8	33	99	1	87.50	Incompatible
Indiana	IN	9	50	100	1	88.89	Incompatible
Massachusetts	MA	9	40	160	1	88.89	Incompatible
California	CA	52	40	80	4	90.00	Incompatible
Washington	WA	10	49	98	1	90.00	Incompatible
Virginia	VA	11	40	100	1	90.91	Incompatible
Michigan	MI	13	38	110	1	92.31	Incompatible
Illinois	IL	17	59	118	1	94.12	Incompatible
Pennsylvania	PA	17	50	203	1	94.12	Incompatible
New York	NY	26	63	150	1	96.15	Incompatible
Texas	TX	38	31	150	1	96.77	Incompatible

Summary. Of the 50 states, 11 are strictly compatible ($\sigma = 0$), 0 fall in the partial range $0 < \sigma < 50$, and 39 are structurally incompatible ($\sigma \geq 50$). No state falls in the intermediate partial range; the distribution is bimodal, clustering at $\sigma = 0$ (single-seat and small-seat states) and $\sigma \geq 50$ (the large-chamber mismatch regime). This bimodality is a mathematical consequence of the GCD structure (Theorem 5), not an accident of the current apportionment.

4.2 Trivially vs. Non-Trivially Compatible States

The 11 strictly compatible states are not a uniform group. A critical distinction separates states whose $\sigma = 0$ carries algorithmic content from those where it is a mathematical artifact:

Trivially compatible ($C = 1$): 7 states. Alaska (AK), Delaware (DE), North Dakota (ND), South Dakota (SD), Vermont (VT), West Virginia (WV), and Wyoming (WY) all have $C = 1$ in the 2020 apportionment. For any state with $C = 1$, we have $\gcd(1, S, H) = 1 = \min(C, S, H)$ regardless of S and H. Strict compatibility ($\sigma = 0$) is therefore a tautology: the single congressional district is the entire state, and every senate and house district is nested inside it by definition. No structural constraint on the state legislative maps follows from this.

Weakly compatible ($C = 2$, $g = 2$): 2 states. Montana (MT, $C = 2$, $S = 50$, $H = 100$, $g = 2$) and New Hampshire (NH, $C = 2$, $S = 24$, $H = 400$, $g = 2$). These states have a trunk of 2 regions — a single bisection of the state. While non-trivial, a 2-region trunk provides only a weak cross-chamber constraint: it amounts to agreeing on a statewide east-west (or similar) boundary between two halves.

Non-trivially compatible ($C \geq 2$, $g = C \geq 3$): 2 states. Alabama (AL, $g = 7$) and Oregon (OR, $g = 6$) are the substantively compatible states. Their senate and house counts are exact multiples of C, yielding a rich multi-region spine that carries genuine algorithmic and geographic content. These two states are the primary motivation for NestSection as a practical redistricting tool.

The headline finding of "11 strictly compatible states" should therefore be read carefully: 7 are trivially compatible ($C = 1$), 2 are weakly compatible ($C = 2$, single bisection), and only 2 — Oregon and Alabama — are meaningfully compatible in the sense that the spine provides a multi-region geographic anchor for all three chambers simultaneously.

4.3 Oregon and Alabama: The Substantive Compatible Cases

Oregon and Alabama are the only US states where the NestSection compatible factorization spine provides genuine multi-region geographic structure.

Oregon: $\gcd(6, 30, 60) = 6$**, trunk** $[2, 3]$**, ratio** $1 : 5 : 10$**.** Oregon's chamber sizes stand in the exact ratio $C : S : H = 6 : 30 : 60 = 1 : 5 : 10$. The compatible spine splits Oregon into $g = 6$ trunk regions, each comprising exactly one congressional district, five senate districts, and ten house districts. The spine trunk $\tau = [2, 3]$ means the algorithm first bisects Oregon into two halves (east/west, following the Cascade Range), then trisects each half into three regions. The resulting 6 trunk regions average roughly 700,000 residents each — comparable to a single congressional district population target. Within each trunk region, the senate map recurses to five districts and the house map to ten, with every house district contained in a senate district and every senate district contained in a congressional district. No boundary zones occur.

Alabama: $\gcd(7, 35, 105) = 7$**, trunk** $[7]$**, ratio** $1 : 5 : 15$**.** Alabama's chamber sizes stand in the exact ratio $C : S : H = 7 : 35 : 105 = 1 : 5 : 15$. The trunk is $\tau = [7]$ — a direct 7-way partition of the state into the 7 congressional districts. Each congressional trunk region contains exactly five senate districts and fifteen house districts. The spine trunk $\tau = [7]$ is a single prime, so the algorithm partitions Alabama directly into 7 regions rather than through a staged bisection as in Oregon. With 4.9 million residents and $C = 7$, each trunk region averages roughly 700,000 residents. The senate subdivides each trunk region into 5 districts ($35/7 = 5$) and the house further into 15 ($105/7 = 15$), with $15/5 = 3$ house districts per senate district.

[Oregon geographic spine: 2-way bisection (east/west) →
3-way trisection of each half → 6 trunk regions (congressional districts).
Each trunk region: 5 senate districts, 10 house districts.
Schematic recursion tree: $[2,3]\|[5]\|[2,5]$*]*

Figure 1: Oregon NestSection geographic spine ($g = 6$, $\tau = [2,3]$). The 6-way top-level split divides Oregon into 6 roughly equal-population regions following the Cascade Range (east-west) and latitudinal divisions (north-south within each half). Each region contains exactly 1 congressional district, 5 senate districts, and 10 house districts. Dashed lines show trunk (congressional) boundaries; solid thin lines show senate boundaries; gray fills show individual house districts.

[Alabama geographic spine: 7-way direct partition → 7 trunk regions
(congressional districts, each $\approx$ 700,000 *residents).*
Each trunk region: 5 senate districts, 15 house districts, 3 house/senate.
Schematic recursion tree: $[7]\|[5]\|[3,5]$*]*

Figure 2: Alabama NestSection geographic spine ($g = 7$, $\tau = [7]$). The 7-way top-level split divides Alabama into 7 roughly equal-population regions following the Black Belt and north-south population corridors. Each region contains exactly 1 congressional district, 5 senate districts, and 15 house districts.

The contrast between Oregon's $\tau = [2,3]$ and Alabama's $\tau = [7]$ is instructive: Oregon's spine decomposes the state via a staged bisection hierarchy (a natural fit for METIS-based recursive bisection), while Alabama's spine requires a direct 7-way partition at the trunk level. Both produce exact nesting with no boundary zones.

4.4 Why No Partially Compatible States?

The gap between strict ($\sigma = 0$) and incompatible ($\sigma \geq 50$) is a structural invariant, proved in Theorem 5: no positive integers (C, S, H) can produce $0 < \sigma < 50$. The empirical absence of such states is therefore an instance of the theorem, not a coincidence of the 2020 apportionment. For σ to fall strictly between 0 and 50, we would need $\gcd(C, S, H)$ to be a divisor of $\min(C, S, H)$ lying strictly between $m/2$ and m; the Bimodality Gap theorem establishes that no such divisor exists.

4.5 Case Study 1: Oregon — Perfect Nesting

Oregon has $C = 6$, $S = 30$, $H = 60$. Then:

$$g = \gcd(6, 30, 60) = 6$$
$$\sigma = (1 - 6/6) \times 100 = 0\%$$

The trunk is the factorization of 6: $[2, 3]$. The compatible spines are:

$$P_C = [2, 3] \quad \text{(product: 6)}$$
$$P_S = [2, 3, 5] \quad \text{(product: 30)}$$
$$P_H = [2, 3, 2, 5] \quad \text{(product: 60)}$$

The geographic interpretation is elegant: bisect Oregon into 2 halves, then trisect each half into 3 regions (6 trunk regions = 6 congressional districts). The senate subdivides each trunk region into 5 senate districts ($30/6 = 5$), and the house further bisects each senate district into 2 house districts ($60/30 = 2$). Every house district is contained in a senate district, every senate district is contained in a congressional district. No boundary zones.

Nesting map. Figure 3 (schematic) illustrates this three-level hierarchy for Oregon. The geographic cuts at each level are determined by GeoSection's ratio-optimal bisection of the state's census graph.

[Oregon NestSection map — 6 trunk regions, 30 senate subdivisions, 60 house leaves — to be generated by redist pipeline]

Figure 3: Schematic NestSection hierarchy for Oregon (C=6, S=30, H=60). Dashed lines show trunk (congressional) boundaries; solid thin lines show senate boundaries; gray fills show individual house districts.

4.6 Case Study 2: North Carolina — Partial Mode 3

North Carolina has $C = 14$, $S = 50$, $H = 120$. Then:

$$g = \gcd(14, 50, 120) = 2$$
$$\sigma = (1 - 2/14) \times 100 \approx 85.7\%$$

The trunk is $[2]$: a single east-west bisection of the state into two halves. The compatible spines are:

$$P_C = [2, 7] \quad \text{(product: 14)}$$
$$P_S = [2, 5, 5] \quad \text{(product: 50)}$$
$$P_H = [2, 3, 4, 5] \quad \text{(product: 120)}$$

Within each half, the congressional map recurses to produce 7 districts; the senate map produces 25 districts; the house map produces 60 districts. The senate and house are far from exact multiples of 7, requiring Mode 3 nesting with boundary tolerance.

Nesting violation. The theoretical mismatch is driven by the tail factorizations [7] (congressional) vs. $[5, 5]$ (senate). In each half, the 7 congressional boundaries do not align with the 5×5 senate recursion. The population fraction in boundary zones is determined empirically by the geographic census-tract distribution; the plan specification reserves $\tau_{\text{pop}} \leq 0.04$ (4% of the state's population).

NestSection map for NC. The NestSection algorithm for NC proceeds as follows:

1. Apply GeoSection to bisect NC into 2 trunk halves.
2. In the western half: apply GeoSection(7) for congressional, GeoSection(25) for senate, NestRefine(25, 60/25) for house.
3. In the eastern half: same recursion.
4. Boundary zones are the census tracts that straddle a congressional/senate boundary; they are assigned to the larger-overlap chamber and flagged.

This guarantees $V_{\text{NC}} \leq 0.04$ while sharing the single east-west trunk.

4.7 Case Study 3: Texas — Structural Incompatibility

Texas has $C = 38$, $S = 31$, $H = 150$. Then:

$$
\begin{aligned}
g &= \gcd(38, 31, 150) = 1 \\
\sigma &= (1 - 1/31) \times 100 \approx 96.8\%
\end{aligned}
$$

The trunk degenerates to the empty sequence $\tau = []$ (the product of no factors is 1): the three chambers share only the state boundary. The compatible spines are simply the independent prime factorizations:

$$
\begin{aligned}
P_C &= [2, 19] \\
P_S &= [31] \quad \text{(prime)} \\
P_H &= [2, 3, 5, 5]
\end{aligned}
$$

No geographic structure can be shared before the individual chamber recursions diverge. NestSection in Mode 3 with TX would require a very large boundary tolerance ($\tau_{\text{pop}} > 0.1$) to force any cross-chamber alignment, effectively making the constraint vacuous.

The fundamental obstacle is the primality of the senate count ($S = 31$): a prime senate chamber is structurally incompatible with any congressional count that is not a multiple of 31. Texas is an extreme case, but the pattern appears in several large states where the senate chamber size was set by historical accident without regard for multi-level divisibility.

4.8 Aggregate Findings

Strict compatibility predictors. The 11 strictly compatible states share two features: (a) most have $C \leq 2$, making $g = C = \min$ trivially by the fact that $g \leq C \leq \min$ when C divides S and H; (b) the multi-seat strictly compatible states (AL, OR) have senate and house counts that are exact multiples of C. Oregon is the most structurally interesting: $C = 6$, $S = 5C$, $H = 10C$, a clean $1 : 5 : 10$ ratio. Alabama has $C = 7$, $S = 5C$, $H = 15C$, a clean $1 : 5 : 15$ ratio.

Score distribution. Among the 39 incompatible states, scores cluster around three plateaus: $\{50\%, 66.7\%, 75\%\}$ for states with small C (2–4 seats) and modest GCDs; $\{80\%–90\%\}$ for mid-size states; and $\{90\%–97\%\}$ for large states where congressional counts are prime or coprime to chamber sizes.

Proposed nestability threshold. A statutory requirement for strict nesting (Mode 1) where $g = \min(C, S, H)$ would apply to 11 states. A weaker requirement of $g \geq 5$ (meaningful shared trunk of at least 5 regions) would apply to only 2 states: Alabama ($g = 7$) and Oregon ($g = 6$). A threshold of $g \geq 3$ would additionally include Louisiana ($g = 3$), Arizona ($g = 3$), New Jersey ($g = 4$), California ($g = 4$), Florida ($g = 4$), Iowa ($g = 2$), Mississippi ($g = 2$), Montana ($g = 2$), North Carolina ($g = 2$), and several others. We revisit this threshold in Section 5.

5 Discussion

5.1 The Legal Argument: Shared Hierarchy as Anti-Gerrymandering

The structural constraint argument. NestSection's central legal contribution is the observation that a shared bisection hierarchy is an anti-gerrymandering mechanism by *construction*, not by detection. Once the trunk is fixed by GeoSection at the spine level, the congressional, senate, and house maps are all constrained to respect the same initial geographic cuts. A partisan actor who wishes to pack a community into a safe congressional district must use the same geographic bisection that the senate and house maps will also use. If the trunk bisects that community, the pack is constrained to one side of the trunk for all three chambers.

This is fundamentally different from post-hoc gerrymandering detection methods (ensemble analysis, partisan symmetry tests), which evaluate a map after it is drawn. NestSection reduces the feasible plan space before any partisan optimization occurs.

Degrees-of-freedom reduction. Formally, the number of distinct geographic configurations compatible with the spine constraint is smaller than the number compatible with three independent maps. In a state with g trunk regions, the congressional map is constrained to respect the g trunk boundaries; only within each trunk region does the congressional mapmaker have freedom. The partisan optimizer's effective search space is reduced by a factor related to the trunk-to-state ratio of geographic complexity.

This is the "mutual constraint" property: because the senate mapmaker is also bound to the same trunk, the congressional mapmaker cannot be instructed to gerrymander in a way that the senate mapmaker could then undo. The constraint is symmetric and enforced at the algorithmic level, not at the political level. It is a process constraint, not a promise that every output map is partisan-neutral.

Comparison to independent maps. With three independent maps, a unified partisan majority can achieve what we call *coherent gerrymandering*: packing or cracking the same geographic community at all three levels simultaneously. NestSection makes coherent gerrymandering structurally harder: a pack at the congressional level propagates to the senate and house levels, reducing the remaining degrees of freedom for optimization at those levels. Conversely, a crack at the congressional level (splitting a community between two congressional districts) automatically splits that community at the senate and house levels in the same geographic location.

Relation to state constitutional nesting requirements. Many state constitutions and redistricting statutes require that districts be "composed of contiguous, compact" territory (e.g., NC Const. art. II, §§3, 5) or that county lines be respected. Some statutes require state house districts to be subsets of senate districts (e.g., Arizona, Indiana). NestSection operationalizes such requirements as a *cross-chamber* computational constraint rather than a within-chamber compactness check.

The NestSection compatibility score σ provides a precise measure of whether such a requirement is feasible: a state with $\sigma = 0$ can satisfy exact nesting without any population imbalance penalty; a state with $\sigma = 96.8\%$ (Texas) cannot satisfy any meaningful nesting without large boundary exceptions.

Elections Clause scope. The Elections Clause (Art. I, §4) grants states authority to regulate the "Times, Places and Manner" of congressional elections, subject to congressional override. A federal statute *requiring* NestSection for congressional maps could face Art. I, §4 challenges from states arguing that the spine constraint intrudes on their redistricting discretion (cf. *Rucho v. Common Cause*, 588 U.S. 684 (2019), which declined to review partisan gerrymandering of congressional maps under federal law, leaving the field to state courts and state law). This paper does not advocate for such a federal mandate.

NestSection's most natural application is to *state legislative* maps — the senate and house tiers — where the governing law is state constitutional provisions on compactness, contiguity, and integrity of political subdivisions, not Art. I, §4. States that voluntarily adopt NestSection for their state legislative chambers, drawing those maps to be consistent with a pre-existing federal congressional map, face no federal constitutional barrier. The compatibility score σ operates on the relationship among all three chamber sizes, but the structural constraint it enforces on the senate and house maps is a matter of state law.

The federal constitutional question arises only in the distinct scenario where Congress were to mandate that state congressional maps conform to a three-chamber spine constraint — a proposal this paper explicitly does not make. The statutory design proposal in this section is addressed to *state legislatures and independent redistricting commissions*, which have plenary authority over state legislative districts and, within Supremacy Clause limits, broad discretion over the manner of drawing congressional districts under Art. I, §4.

Statutory design: the $g \geq k$ threshold. We propose a two-tier statutory structure:

1. **Mandatory strict nesting** for states where $\gcd(C, S, H) \geq 5$: Oregon ($g = 6$) and Alabama ($g = 7$) currently qualify. These states can achieve exact hierarchical nesting (Mode 1) with no population tolerance. A statute requiring Mode 1 for these states would be straightforwardly achievable with existing redistricting algorithms.

2. **Best-effort nesting** (Mode 3 with $\tau_{\text{pop}} \leq 0.02$) for states where $\gcd(C, S, H) \geq 2$: 22 states currently qualify. The statute would require the published boundary-zone fraction τ to be minimized and publicly reported, creating an empirically verifiable standard for cross-chamber consistency.

Judicial enforceability. Judicial enforceability of a NestSection requirement raises two distinct questions. First, who has standing to challenge a non-nested map? The most natural plaintiffs are political subdivisions (counties, municipalities) whose territories span chamber boundaries inconsistently — they can argue that the inconsistent boundaries create administrative burdens that serve no compelling interest. Second, what is the standard of review? If NestSection is framed as a procedural rule (the map must be derived from a mathematical process), courts reviewing it apply rational-basis scrutiny, not strict scrutiny — the bar is low. A statute stating “state legislative districts shall be derived by refining the congressional factorization tree” has a rational basis in administrative consistency and resistance to manipulation. Courts striking down such a requirement would need to find it facially arbitrary, which the Bimodality Gap Theorem makes difficult: the compatible states are a mathematically defined class, not a politically selected one.

5.2 Limitations

Incompatible states. For 18 states with $g = 1$ and congressional seat counts $C \geq 5$, NestSection reduces to three independent GeoSection runs with a shared census graph. The algorithm provides no geographic anchor beyond the state boundary. Statutory mandates for cross-chamber consistency in these states would require either (a) changing the senate chamber size to be compatible with C (a major political change), or (b) tolerating large boundary zones that may undermine the legal defensibility of the nesting constraint.

Population balance. The spine constraint interacts with the statutory $\pm 0.5\%$ population equality requirement for congressional districts. In a strict-nesting state, the trunk regions must have populations that are simultaneously divisible into C/g, S/g, and H/g equal-population sub-districts. Population heterogeneity within trunk regions may require slight adjustments to trunk boundaries (Mode 2 or Mode 3) even in nominally compatible states. The empirical validation in Section 4 is theoretical; full empirical confirmation requires running the NestSection pipeline on actual census data.

Single-seat states. Eleven states have $C = 1$, which trivially achieves $g = 1 = \min(C, S, H)$ and $\sigma = 0$. However, a single-seat congressional delegation has no internal geographic structure to share with the state legislature. The "strict compatibility" of these states is a mathematical artifact of the formula rather than a meaningful cross-chamber constraint. In practice, single-seat states require the congressional district to be the entire state, which makes any cross-chamber nesting trivial (the congressional district contains all senate and house districts by definition).

Nebraska's unicameral legislature. Nebraska is unique among US states in having a unicameral legislature (49 senators, no separate house). The nest.rs implementation uses $H = S = 49$ for Nebraska; in practice, NestSection would apply only to the two-chamber relationship (congressional vs. senate), and the house tier is vacuous.

Senate-first vs. congressional-first recursion. NestSection as defined uses the GCD trunk and then recurses independently within each trunk region for all three chambers. An alternative is to define the spine from the largest chamber (house) and coarsen upward. This is the Mode 2 design from the plan: using the senate as the spine and placing congressional seats at coarsened senate groups. The Mode 2 approach is mathematically equivalent to Mode 1 when divisibility holds, but differs in the within-region recursion order when it does not. We defer a full comparison to future work.

Approximate compatibility: Arizona and Indiana as illustrative cases. Two near-compatible states illustrate a continuum that the strict bimodality threshold ($\sigma = 0$ vs. $\sigma \geq 50\%$) obscures. Arizona has $C = 9$, $S = 30$, $H = 60$, giving $\gcd(9, 30, 60) = 3$ and $\sigma = 67\%$. The spine trunk [3] would create a three-way top-level geographic split shared across all chambers — a natural east/west/central Arizona division (roughly: Phoenix metro, Tucson corridor, and rural east). Each side would hold 3 congressional, 10 senate, and 20 house districts. This is a substantive, non-trivial compatible structure with real geographic content, even at $\sigma = 67\%$ (incompatible by strict definition). It illustrates that the strict bimodality threshold hides a continuum of approximate compatibility that future work could formalise.

Indiana has $C = 9$, $S = 50$, $H = 100$, giving $\gcd(9, 50, 100) = 1$ and $\sigma = 89\%$. This is the hardest incompatible case: 9 congressional, 50 senate, and 100 house districts share no common factor. Any NestSection spine for Indiana would require either (a) rounding congressional districts to 10 (impossible by apportionment), or (b) accepting that state legislative maps cannot be nested in the congressional structure. Indiana represents the class of states where NestSection applies only to the state legislative chambers internally — senate nests into house: $\gcd(50, 100) = 50$, perfect compatibility — without congressional linkage.

5.3 Future Work

Empirical pipeline validation. The most important near-term extension is running NestSection on actual census data for the case-study states (Oregon, North Carolina, Texas) and measuring: (a) the actual nesting violation V achieved on census geography; (b) the edge-cut premium vs. independent GeoSection runs; (c) the partisan outcome distribution across 100 GeoSection seeds.

Gerrymander resistance hypothesis. Section 1 conjectures that NestSection reduces partisan variance by approximately 30–50% in moderate-compatibility states. Testing this hypothesis requires the ensemble-diagnostics infrastructure from `redist-analysis::ensemble_diagnostics` (R-hat, ESS, Hamming autocorrelation). A formal test would compare the standard deviation of Democratic seat share across 100 NestSection seeds vs. 100 independent GeoSection seeds for NC and WI.

Reapportionment stability. When a state gains or loses a congressional seat after reapportionment, the GCD g may change, altering which nesting mode applies. Montana gained a seat in 2020 (going from $C = 1$ to $C = 2$), preserving $\sigma = 0$ because $g = 2$ still equals min $= 2$. Texas will likely gain seats after 2030, possibly achieving a GCD > 1 with its senate. A longitudinal analysis of how σ has changed across census decades would quantify the stability of nesting constraints over reapportionment cycles.

VRA interaction. The NestSection trunk constraint interacts with Voting Rights Act Section 2 compliance. If a minority-majority community is bisected by the trunk, it will be split at all three legislative levels simultaneously. This could harm VRA compliance if the split reduces the minority's ability to elect representatives of their choice. The `redist analyze -types bloc-voting` Callais pipeline provides the tools to measure this interaction. A NestSection implementation should include a VRA preflight check (analogous to the `callais_preflight` function in the existing pipeline) to detect cases where the spine bisects a protected community.

6 Conclusion

We introduced NestSection, an algorithm for constructing hierarchically consistent redistricting maps across the congressional, state senate, and state house chambers. The central contribution is the CompatibleSpines algorithm, which identifies the largest geographic structure that all three chambers can share: the factorization trunk determined by $g = \gcd(C, S, H)$.

The spine compatibility score $\sigma(C, S, H) = (1 - g/\min(C, S, H)) \times 100$ provides a single number summarizing how tightly the three chambers are structurally linked. We applied this score to all 50 US states under the 2020 apportionment, finding that 11 states admit strict nesting ($\sigma = 0$) and 39 require best-effort Mode 3 nesting ($\sigma \geq 50$). The absence of states in the intermediate range is a structural property of current US apportionments, not an artifact of the metric.

The three case studies illustrate the range of outcomes:

- **Oregon** ($g = 6$, $\sigma = 0$): A clean $1 : 5 : 10$ congressional/senate/house ratio admits exact hierarchical nesting. The three chambers share a 6-region trunk and diverge only in the tail splits within each trunk region.

- **North Carolina** ($g = 2$, $\sigma = 85.7\%$): A single trunk bisection is all that can be shared. Mode 3 nesting with $\tau_{\text{pop}} \leq 0.04$ is achievable, but the partisan consequences of the trunk bisection are non-trivial.
- **Texas** ($g = 1$, $\sigma = 96.8\%$): The primality of the 31-seat senate makes the state structurally incompatible. No meaningful cross-chamber nesting is achievable without restructuring the chamber sizes.

The legal argument is constructive: requiring NestSection for compatible states reduces the degrees of freedom available for partisan optimization before the map is drawn, not after. States with $g \geq 2$ (22 states) could be subject to a best-effort nesting mandate; states with $g \geq 5$ (Oregon and Alabama) could be subject to a strict nesting mandate with zero boundary tolerance. Such a statute would be the first anti-gerrymandering rule that operates at the level of geometric structure rather than partisan outcome.

The implementation is available in `redist-apportion::nest` and is fully tested against all 50 states. Future work will validate the algorithm on actual census geography and measure the gerrymander-resistance effect empirically.

Acknowledgments. The authors thank the NCSL for chamber-size data and the Census Bureau for the 2020 apportionment figures. The `bisect` Rust workspace is available at `https://github.com/apportionment-research/redist`.

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